

Lecture Six

1 Vector Space

1.1 Basic Concepts and Examples

Definition 1.1.1 A vector space is a non empty set V of objects, called vectors, on which are defined two operations called addition and multiplication by scalars (real numbers) subject to ten axioms (or rules) listed below. The axioms must hold for all vectors u, v and w in V and for all scalars c and d .

Addition

1. $u + v \in V$ (closure under addition).
2. $u + v = v + u$ (commutative property)
3. $(u + v) + w = u + (v + w)$ (Associative property)
4. V has a zero vector 0 such that for every $u \in V$, $u + 0 = u$.
5. For every u in V , there is a vector in V denoted by $(-u)$ such that $u + (-u) = 0$

Scalar Multiplication

6. cu in V (Closure under scalar multiplication)
7. $c(u + v) = cu + cv$ (Distributive property)
8. $(c + d)u = cu + du$
9. $c(du) = (cd)u$ (Associative)
10. $1.u = u$ (Scalar identity)

Example 1.1.1 $\mathbb{R}^2$ with the standard operations is a vector space.

Proof

$\forall u, v, w \in \mathbb{R}^2$ and $\forall c, d \in \mathbb{R}$, firstly we can define the vectors as follows

$$\begin{aligned} u &= (u_1, u_2) \quad ; u_i \in \mathbb{R} \\ v &= (v_1, v_2) \quad ; v_i \in \mathbb{R} \\ w &= (w_1, w_2) \quad ; w_i \in \mathbb{R} \end{aligned}$$

1.

$$v + u = (v_1 + u_1, v_2 + u_2) \in \mathbb{R}^2$$

2.

$$\begin{aligned} (v + u) + w &= (v_1 + u_1, v_2 + u_2) + (w_1, w_2) = (v_1 + u_1 + w_1, v_2 + u_2 + w_2) \\ &= (v_1 + (u_1 + w_1), v_2 + (u_2 + w_2)) = (v_1, v_2) + (u_1 + w_1, u_2 + w_2) \\ &= v + (u + w) \end{aligned}$$

3.

$$\begin{aligned} u + v &= (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) = (v_1 + u_1, v_2 + u_2) = (v_1, v_2) + (u_1, u_2) \\ &= v + u \end{aligned}$$

4.

$$\exists (0, 0) \in \mathbb{R}^2 ; v + 0 = (v_1, v_2) + (0, 0) = (v_1, v_2) = v$$

5.

To prove that the algebraic structure $(\mathbb{R}^2, +, \cdot)$ forms a vector space over the field of real numbers $\mathbb{R}$, we must verify the ten fundamental axioms.

Let the standard operations on $\mathbb{R}^2$ be defined as follows for any vectors $\mathbf{u} = (u_1, u_2)$, $\mathbf{v} = (v_1, v_2)$, $\mathbf{w} = (w_1, w_2) \in \mathbb{R}^2$ and scalars $c, d \in \mathbb{R}$:

$$\text{Vector Addition: } \mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2)$$

$$\text{Scalar Multiplication: } c \cdot \mathbf{u} = (cu_1, cu_2)$$

1. Axiom 1: Closure under Addition

Since $u_1, v_1 \in \mathbb{R}$ and $u_2, v_2 \in \mathbb{R}$, their sums $(u_1 + v_1)$ and $(u_2 + v_2)$ are also real numbers by the closure property of $\mathbb{R}$. Therefore:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2) \in \mathbb{R}^2$$

2. Axiom 2: Commutativity of Addition

Using the commutative property of real number addition ($u_1 + v_1 = v_1 + u_1$):

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2) = (v_1 + u_1, v_2 + u_2) = \mathbf{v} + \mathbf{u}$$

3. Axiom 3: Associativity of Addition

Using the associative property of real numbers:

$$\begin{aligned} (\mathbf{u} + \mathbf{v}) + \mathbf{w} &= ((u_1 + v_1) + w_1, (u_2 + v_2) + w_2) \\ &= (u_1 + (v_1 + w_1), u_2 + (v_2 + w_2)) = \mathbf{u} + (\mathbf{v} + \mathbf{w}) \end{aligned}$$

4. Axiom 4: Existence of Additive Identity

There exists a zero vector $\mathbf{0} = (0, 0) \in \mathbb{R}^2$ such that:

$$\mathbf{u} + \mathbf{0} = (u_1 + 0, u_2 + 0) = (u_1, u_2) = \mathbf{u}$$

5. Axiom 5: Existence of Additive Inverse

For every vector $\mathbf{u} = (u_1, u_2) \in \mathbb{R}^2$, there exists an inverse $-\mathbf{u} = (-u_1, -u_2) \in \mathbb{R}^2$ such that:

$$\mathbf{u} + (-\mathbf{u}) = (u_1 + (-u_1), u_2 + (-u_2)) = (0, 0) = \mathbf{0}$$

6. Axiom 6: Closure under Scalar Multiplication

Since $c, u_1, u_2 \in \mathbb{R}$, the products cu_1 and cu_2 are real numbers. Thus:

$$c \cdot \mathbf{u} = (cu_1, cu_2) \in \mathbb{R}^2$$

7. Axiom 7: Distributivity of Scalar Multiplication over Vector Addition

Applying the distributive law of real numbers:

$$\begin{aligned} c \cdot (\mathbf{u} + \mathbf{v}) &= c \cdot (u_1 + v_1, u_2 + v_2) = (c(u_1 + v_1), c(u_2 + v_2)) \\ &= (cu_1 + cv_1, cu_2 + cv_2) = c \cdot \mathbf{u} + c \cdot \mathbf{v} \end{aligned}$$

8. Axiom 8: Distributivity of Scalar Multiplication over Scalar Addition

Applying the distributive law of real numbers:

$$\begin{aligned} (c + d) \cdot \mathbf{u} &= ((c + d)u_1, (c + d)u_2) = (cu_1 + du_1, cu_2 + du_2) \\ &= (cu_1, cu_2) + (du_1, du_2) = c \cdot \mathbf{u} + d \cdot \mathbf{u} \end{aligned}$$

9. Axiom 9: Associativity of Scalar Multiplication

Using the associative property of real multiplication:

$$c \cdot (d \cdot \mathbf{u}) = c \cdot (du_1, du_2) = (c(du_1), c(du_2)) = ((cd)u_1, ((cd)u_2)) = (cd) \cdot \mathbf{u}$$

10. Axiom 10: Scalar Identity Property

For the real multiplicative identity $1 \in \mathbb{R}$:

$$1 \cdot \mathbf{u} = (1 \cdot u_1, 1 \cdot u_2) = (u_1, u_2) = \mathbf{u}$$

Conclusion: Since all ten axioms hold, $(\mathbb{R}^2, +, \cdot)$ is a valid Vector Space. ■

Example 1.1.2 *Summary of some important vector spaces.*

1. $\mathbb{R}^n$ – All vectors with n real components.
2. $\mathbb{C}^n$ – All vectors with n complex components.
3. Matrix space $M_{m \times n}(\mathbb{R})$ – All $m \times n$ matrices with real entries.
4. Matrix space $M_{m \times n}(\mathbb{C})$ – All $m \times n$ matrices with complex entries.
5. Continuous functions $C([a, b], \mathbb{R})$ – All real-valued continuous functions on $[a, b]$.
6. Differentiable functions $C^1([a, b], \mathbb{R})$ – Real-valued functions with continuous derivatives on $[a, b]$.

Example 1.1.3 *Prove that $V = \{(x, y) \in \mathbb{R}^2 : y = mx\}$, m fixed, is a vector space.*

Proof

Let $u = (x_1, mx_1)$, $v = (x_2, mx_2) \in V$, and $c \in \mathbb{R}$.

1. **Closure under addition:**

$$u + v = (x_1, mx_1) + (x_2, mx_2) = (x_1 + x_2, mx_1 + mx_2) = (x_1 + x_2, m(x_1 + x_2)) \in V$$

2. **Closure under scalar multiplication:**

$$cu = c(x_1, mx_1) = (cx_1, cmx_1) = (cx_1, m(cx_1)) \in V$$

3. **Commutativity of addition:**

$$u + v = (x_1 + x_2, m(x_1 + x_2)) = (x_2 + x_1, m(x_2 + x_1)) = v + u$$

4. **Associativity of addition:** Let $w = (x_3, mx_3) \in V$. Then

$$(u + v) + w = ((x_1 + x_2) + x_3, m((x_1 + x_2) + x_3)) = (x_1 + (x_2 + x_3), m(x_1 + (x_2 + x_3))) = u + (v + w)$$

5. **Existence of additive identity:**

$$0_V = (0, 0) \in V, \quad u + 0_V = (x_1 + 0, mx_1 + 0) = (x_1, mx_1) = u$$

6. **Existence of additive inverse:**

$$-u = (-x_1, -mx_1) \in V, \quad u + (-u) = (x_1 - x_1, mx_1 - mx_1) = (0, 0) = 0_V$$

7. **Distributivity of scalar multiplication over vector addition:**

$$c(u + v) = c(x_1 + x_2, m(x_1 + x_2)) = (c(x_1 + x_2), mc(x_1 + x_2)) = (cx_1 + cx_2, m(cx_1 + cx_2)) = cu + cv$$

8. **Distributivity of scalar addition over scalar multiplication:**

$$(c + d)u = ((c + d)x_1, m((c + d)x_1)) = (cx_1 + dx_1, m(cx_1 + dx_1)) = cu + du$$

9. **Associativity of scalar multiplication:**

$$c(du) = c(dx_1, mdx_1) = (cdx_1, mcdx_1) = (cd)u$$

10. **Multiplicative identity:**

$$1 \cdot u = (1 \cdot x_1, 1 \cdot mx_1) = (x_1, mx_1) = u$$

Therefore, V satisfies all 10 axioms of a vector space over $\mathbb{R}$:

V is a vector space.

1.2 Subspaces

Definition 1.2.2 Let V be a vector space over a field F . A nonempty subset H of V is called a **subspace** of V if H is itself a vector space under the same vector addition and scalar multiplication defined on V .

Equivalently, H is a subspace of V if:

1. $\mathbf{0} \in H$,
2. For all $u, v \in H$, we have $u + v \in H$,
3. For all $u \in H$ and $c \in F$, we have $cu \in H$.

Example 1.2.4 Some examples of Subspaces:

1. $\{(x, y) \in \mathbb{R}^2 : y = mx\}$, where m is fixed.

2.

$$H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x + y + z = 0 \right\}.$$

3.

$$H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^3 : x = y \right\}.$$

4. The xy -plane:

$$H = \left\{ \begin{bmatrix} x \\ y \\ 0 \end{bmatrix} : x, y \in \mathbb{R} \right\}.$$

5. The x -axis:

$$H = \left\{ \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} : x \in \mathbb{R} \right\}.$$

6. The set of all symmetric matrices:

$$H = \{A \in M_n(\mathbb{R}) : A^T = A\}.$$

7. The set of all diagonal matrices:

$$H = \{A \in M_n(\mathbb{R}) : A \text{ is diagonal}\}.$$

8. The set of all polynomials of degree at most n :

$$P_n = \{a_0 + a_1x + \cdots + a_nx^n\}.$$

9. The set of all continuous functions on $[a, b]$:

$$C[a, b].$$

10. The zero subspace:

$$H = \{\mathbf{0}\}.$$

Example 1.2.5 A plane in $\mathbb{R}^3$ not through the origin is not a subspace of $\mathbb{R}^3$ because the plane does not contain the zero vector of $\mathbb{R}^3$. Similarly, a line in $\mathbb{R}^2$ not through the origin is not a subspace of $\mathbb{R}^2$.

The next example illustrates one of the most common ways of describing a subspace. The term linear combination refers to any sum of scalar multiples of vectors and $\text{span}\{v_1, \dots, v_p\}$ denotes the set of all vectors that can be written as linear combination of $v_1, \dots, v_p$.

Example 1.2.6 Let V be a vector space and $H = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \{c_1\mathbf{v}_1 + c_2\mathbf{v}_2 \mid c_1, c_2 \in \mathbb{R}\}$. To prove that H is a subspace of V , we must show it satisfies the three subspace criteria:

1. **The Zero Vector:** Choosing $c_1 = 0$ and $c_2 = 0$, we have:

$$\mathbf{0} = 0\mathbf{v}_1 + 0\mathbf{v}_2$$

Since $\mathbf{0}$ can be expressed as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$, then $\mathbf{0} \in H$.

2. **Closure under Addition:** Let $\mathbf{u}, \mathbf{w} \in H$. Then there exist scalars c_1, c_2, d_1, d_2 such that:

$$\mathbf{u} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 \quad \text{and} \quad \mathbf{w} = d_1\mathbf{v}_1 + d_2\mathbf{v}_2$$

Adding them:

$$\mathbf{u} + \mathbf{w} = (c_1 + d_1)\mathbf{v}_1 + (c_2 + d_2)\mathbf{v}_2$$

Since $(c_1 + d_1)$ and $(c_2 + d_2)$ are scalars, $\mathbf{u} + \mathbf{w} \in H$.

3. **Closure under Scalar Multiplication:** Let $\mathbf{u} \in H$ and $k \in \mathbb{R}$. Then:

$$k\mathbf{u} = k(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) = (kc_1)\mathbf{v}_1 + (kc_2)\mathbf{v}_2$$

Since (kc_1) and (kc_2) are scalars, $k\mathbf{u} \in H$.

Conclusion: Since H satisfies all three conditions, H is a subspace of V .

Theorem 1.2.1 If $v_1, \dots, v_p$ are in a vector space, then $\text{span}\{v_1, \dots, v_p\}$ is a subspace of V .

Example 1.2.7 Let H be the set of all vectors of the form $(a - 3b, b - a, a, b)$ where a and b are arbitrary scalars. That $H = (a - 3b, b - a, a, b) : a, b \in \mathbb{R}$ Show that H is a subspace of $\mathbb{R}^4$

Proof

First, we express H as a linear combination:

$$\begin{bmatrix} a - 3b \\ b - a \\ a \\ b \end{bmatrix} = a \begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} -3 \\ 1 \\ 0 \\ 1 \end{bmatrix}$$

Let $\mathbf{v}_1 = [1, -1, 1, 0]^T$ and $\mathbf{v}_2 = [-3, 1, 0, 1]^T$. Thus, $H = \text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.

- Zero Vector:** For $a = 0, b = 0$, the vector is $\mathbf{0} \in H$.
- Addition:** Given $\mathbf{u} = a_1\mathbf{v}_1 + b_1\mathbf{v}_2$ and $\mathbf{w} = a_2\mathbf{v}_1 + b_2\mathbf{v}_2$, then $\mathbf{u} + \mathbf{w} = (a_1 + a_2)\mathbf{v}_1 + (b_1 + b_2)\mathbf{v}_2 \in H$.
- Scalar Multiplication:** For $k \in \mathbb{R}$, $k(a\mathbf{v}_1 + b\mathbf{v}_2) = (ka)\mathbf{v}_1 + (kb)\mathbf{v}_2 \in H$.

Conclusion: H satisfies all criteria and is a subspace of $\mathbb{R}^4$.

Example 1.2.8 For what values of h , y will be in the subspace of $\mathbb{R}^3$ spanned by v_1, v_2, v_3 if $v_1 = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}$, $v_2 =$

$$\begin{bmatrix} 5 \\ -4 \\ -7 \end{bmatrix}, v_3 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix} \text{ and } y = \begin{bmatrix} -4 \\ 3 \\ h \end{bmatrix}$$

Sol

To determine if $\mathbf{y}$ is in the span, we solve the augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ -1 & -4 & 1 & 3 \\ -2 & -7 & 0 & h \end{array} \right] \xrightarrow[R_3 \rightarrow R_3 + 2R_1]{R_2 \rightarrow R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 3 & -6 & h - 8 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 3R_2} \left[\begin{array}{ccc|c} 1 & 5 & -3 & -4 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & h - 5 \end{array} \right]$$

The system is consistent if $h - 5 = 0$. Therefore, $\mathbf{y}$ is in the subspace for $h = 5$.

Exercise 1.2.1 Answer the following:

- Let V be the first quadrant in the xy -plane

$$V = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} ; x \geq 0, y \geq 0 \right\}$$

- (a) If u and v are in V , is $u + v$ in V ? why?
- (b) Find a specific vector u in V and a specific scalar c such that cu is not in V . (This is enough to show that V is not a vector space).

2. Let W be the union of the first and third quadrants in the xy -plane

$$w = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : xy \geq 0 \right\}$$

- (a) If u is in w and c is any scalar, is cu in w ? why?
- (b) Find specific vectors u and v in w such that $u + v$ is not in w .

1.3 The Null Space

Consider the following system of homogeneous equations:

$$\begin{array}{rrrrr} x_1 & - & 3x_2 & - & 2x_3 & = & 0 \\ -5x_1 & + & 9x_2 & + & x_3 & = & 0 \end{array} \quad (1.3.1)$$

In matrix form, this system is written as $Ax = 0$, where

$$\begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$$

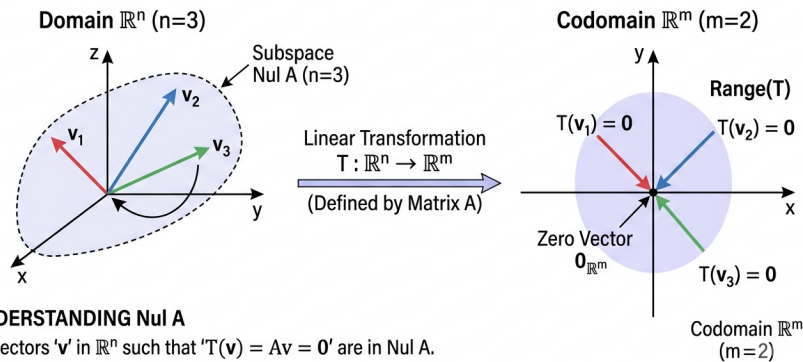
Recall that the set of all x that satisfy 1.3.1 is called the solution set of the system, often it is convenient to relate this set directly to the matrix A and the equation $Ax = 0$. We call the set of x that satisfy $Ax = 0$ the **null space** of the matrix A .

Definition 1.3.3 The **null space** of an $m \times n$ matrix A , written as $\text{Nul}(A)$ is the set of all solutions of the homogeneous equation $Ax = 0$. In set notations

$$\text{Nul}A = \{x : x \text{ is in } \mathbb{R}^n\} \text{ and } Ax = 0$$

$\text{Nul}A$ is the set of all x in $\mathbb{R}^n$ that are mapped into the zero vector of $\mathbb{R}^m$ via the linear transformation $x \rightarrow Ax$.

ILLUSTRATING THE CONCEPT OF Nul A (NULL SPACE)



Example 1.3.9 Let $A = \begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix}$ and let $u = \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix}$. Determine if u belongs to null space of A .

Sol

It means that we will test that if u satisfy $Au = 0$. Simply compute

$$\begin{bmatrix} 1 & -3 & -2 \\ -5 & 9 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} 5 - 9 + 4 \\ -25 + 27 - 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus u is in $\text{Nul}A$.

Theorem 1.3.2 The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$. Equivalently the set of all solutions to a system $Ax = 0$ of an homogeneous linear equations in n unknowns is a subspace of $\mathbb{R}^n$.

Example 1.3.10 Find a spanning set for the null space of the matrix

$$A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}$$

Sol

The first step is to find the general solution of $Ax = 0$ in terms of free variables. Row reduce the augmented matrix $[A \ 0]$ to reduced echelon form in order to write the basic variables in terms of the free variable

$$\begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ -3 & 6 & -1 & 1 & -7 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 2 & 3 & -1 \\ 0 & 0 & 5 & 10 & -10 \\ 0 & 0 & 1 & 2 & -2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & 2 & -2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Now

$$\begin{aligned} x_1 - 2x_2 - x_4 + 3x_5 &= 0 \\ x_2 + 2x_4 - 2x_5 &= 0 \\ 0 &= 0 \end{aligned}$$

The general solution is $x_1 = 2x_2 + x_4 - 3x_5$ with x_2, x_4 and x_5 free. Now, decompose the vector giving the general solution into a linear combination of vectors where the weights are the free variables. That is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2x_2 + x_4 - 3x_5 \\ x_2 \\ -2x_4 + 2x_5 \\ x_4 \\ x_5 \end{bmatrix} = x_2 \underbrace{\begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}}_u + x_4 \underbrace{\begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \\ 0 \end{bmatrix}}_v + x_5 \underbrace{\begin{bmatrix} -3 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix}}_w = x_2u + x_4v + x_5w$$

$\Rightarrow$ every linear combination is an element of $\text{Nul } A$ and vice versa. Thus $\{u, v, w\}$ is a spanning set for $\text{Nul } A$.

1.4 The Column Space

Another important subspace associated with a matrix is its column space. Unlike the null space, the column space is defined explicitly via linear combinations.

Definition 1.4.4 The **column space** of an $m \times n$ matrix A , written as $\text{Col } A$, is the set of all linear combinations of the columns of A . If $A = [a_1 \ \dots \ a_n]$, then

$$\text{Col } A = \text{span}\{a_1, \dots, a_n\}$$

Note that

1. A typical vector in $\text{Col } A$ is also can be written as Ax for some x because the notation Ax stands for linear combination of the columns of A . That is

$$\boxed{\text{Col } A = \{b : b = Ax \text{ for some } x \text{ in } \mathbb{R}^n\}}$$

2. The notation Ax for vectors in $\text{Col } A$ also shows that $\text{Col } A$ is the range of the linear transformation $x \rightarrow Ax$

Example 1.4.11 Find a matrix A such that $W = \text{Col } A$

$$W = \left\{ \begin{bmatrix} 6a - b \\ a + b \\ -7a \end{bmatrix} : a, b \in \mathbb{R} \right\}$$

Sol

First, write W as a set of linear combinations.

$$W = \left\{ a \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix} + b \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} : a, b \in \mathbb{R} \right\} = \text{span} \left\{ \begin{bmatrix} 6 \\ 1 \\ -7 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}$$

Second, use the vectors in the spanning set as the columns of A .

$$A = \begin{bmatrix} 6 & -1 \\ 1 & 1 \\ -7 & 0 \end{bmatrix}$$

Theorem 1.4.3 The column space of an $m \times n$ matrix A is all of $\mathbb{R}^m$ if and only if the equation $Ax = b$ has a solution for each b in $\mathbb{R}^m$.

Example 1.4.12 Let

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix}$$

- (a) If the column space of A is a subspace of $\mathbb{R}^k$, what is k ?
 (b) If the null space of A is a subspace of $\mathbb{R}^k$, what is k ?

Sol

- (a) The columns of A each have three entries, so $\text{Col } A$ is a subspace of $\mathbb{R}^k$ where $k = 3$.
 (b) A vector space x such that Ax is defined must have four entries, so $\text{Nul } A$ is a subspace of $\mathbb{R}^k$ where $k = 4$.

Example 1.4.13 With A as in example 1.4.12, let $u = \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix}$ and $v = \begin{bmatrix} 3 \\ -1 \\ 3 \end{bmatrix}$.

- Determine if u is in $\text{Nul } A$. Could u be in $\text{Col } A$?
- Determine if v is in $\text{Col } A$. Could v be in $\text{Nul } A$?

Sol

- Simply compute the product Au .

$$Au = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \\ 3 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Obviously, u is not a solution of $Ax = 0$, so u is not in $\text{Nul } A$. Also, with four entries, u could not possibly be in $\text{Col } A$, since $\text{Col } A$ is a subspace of $\mathbb{R}^3$.

- Reduce $[A \ v]$ to an echelon form.

$$[A \ v] = \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ -2 & -5 & 7 & 3 & -1 \\ 3 & 7 & -8 & 6 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 4 & -2 & 1 & 3 \\ 0 & 1 & -5 & -4 & -2 \\ 0 & 0 & 0 & 17 & 1 \end{bmatrix}$$

At this point, it is clear that the equation $Ax = v$ is consistent, so v is in $\text{Col } A$. With only three entries, v could not possibly be in $\text{Nul } A$, since $\text{Nul } A$ is a subspace of $\mathbb{R}^4$.

Exercise 1.4.2 Solve the following

Determine if $\mathbf{w} = \begin{bmatrix} 5 \\ -3 \\ 2 \end{bmatrix}$ is in $\text{Nul } A$, where

$$A = \begin{bmatrix} 5 & 21 & 19 \\ 13 & 23 & 2 \\ 8 & 14 & 1 \end{bmatrix}.$$

Determine if $\mathbf{w} = \begin{bmatrix} 1 \\ 3 \\ -4 \end{bmatrix}$ is in $\text{Nul } A$, where

$$A = \begin{bmatrix} 3 & -5 & -3 \\ 6 & -2 & 0 \\ -8 & 4 & 1 \end{bmatrix}.$$

Let $A = \begin{bmatrix} -6 & 12 \\ -3 & 6 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$. Determine if $\mathbf{w}$ is in $\text{Col } A$. Is $\mathbf{w}$ in $\text{Nul } A$?

Let $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$ and $\mathbf{w} = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Determine if $\mathbf{w}$ is in $\text{Col } A$. Is $\mathbf{w}$ in $\text{Nul } A$?

Find A such that the given set is $\text{Col } A$.

$$\left\{ \begin{bmatrix} 2s + 3t \\ r + s - 2t \\ 4r + s \\ 3r - s - t \end{bmatrix} : r, s, t \text{ real} \right\}$$

$$\left\{ \begin{bmatrix} b - c \\ 2b + c + d \\ 5c - 4d \\ d \end{bmatrix} : b, c, d \text{ real} \right\}$$